

Iteration 7: Zero-Shot Multilingual NLI Classification

4 manually labelled language
datasets (EN / DE / SV / NL)
Combined corpus: 33,878 incident records

Binary label mapping:
`CASE_TYPE` = “Process Safety” $\rightarrow$ 1, else 0
No supervised training, no few-shot exemplars

joeddav/xlm-roberta-large-xnli
Multilingual NLI-adapted transformer

Descriptive candidate hypotheses
 c_1 : “*Process safety incident
involving equipment failure,
leak, explosion, shutdown, fire, \dots*”
 c_2 : “*Non-process safety in-
cident involving personal
injury, slip, fall, vehicle accident, \dots*”

Zero-shot entailment inference
 $\hat{y} = \arg \max_{c \in \mathcal{C}} P(\text{entail} \mid x, c)$
HuggingFace
zero-shot-classification pipeline
(CUDA / MPS / CPU),
checkpoint every 200 records

Evaluation: per-language
and combined multilingual
Accuracy, PS P/R/F1, Non-PS F1, Macro-F1,
confusion matrices, ROC, PR curves